

Research Interests

Diffusion language models, AI safety and interpretability, formal methods.

Work Experience

- 2025/08 – Present ♦ **The University of Texas at Austin, Postdoctoral Fellow**
Advisors: Sanjay Shakkottai and Adam Klivans
Institute for Foundations of Machine Learning

Education

- 2019/08 – 2025/08 ♦ **University of Pennsylvania, Ph.D. in Computer Science**
Advisors: Rameev Alur and Eric Wong
Thesis: Trustworthy Machine Learning: Specification, Verification, and Explanation
- 2015/08 – 2019/05 ♦ **Yale University, B.S. in Computer Science and Mathematics**
Advisor: Ruzica Piskac
Double major, undergraduate research on program analysis and Haskell.

Publications

Preprints

- 1 Alan Li, Rahul Saha, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Praveesh K Kothari, and Raghu Meka, “The Grothendieck Constant is Less Than $\frac{\pi}{2 \log(1+\sqrt{2})} - 10^{-5}$ ”, *arXiv preprint arXiv:2606.03991*, 2026.
- 2 **Anton Xue***, Litu Rout*, Constantine Caramanis, and Sanjay Shakkottai, “AnCoder: Anchored Code Generation via Discrete Diffusion Models”, *arXiv preprint arXiv:2602.17688*, 2026.
- 3 Shreya Havaladar*, Weiqiu You*, Chaehyeon Kim, **Anton Xue**, Helen Jin, Marco Gatti, Bhuvnesh Jain, Helen Qu, Amin Madani, Daniel A Hashimoto, Gary E Weissman, Rajat Deo, Sameed Khatana, Lyle Ungar, and Eric Wong, “T-FIX: Text-Based Explanations with Features Interpretable to eXperts”, *arXiv preprint arXiv:2511.04070*, 2025.

Conferences and Journals

- 1 Shailesh Sridhar, **Anton Xue**, and Eric Wong, “Missingness Bias Calibration in Feature Attribution Explanations”, in *International Conference on Learning Representations (ICLR)*, 2026.
- 2 Andy Yang, Christopher Watson, **Anton Xue**, Satwik Bhattamishra, Jose Llarena, William Merrill, Emile Dos Santos Ferreira, Anej Svete, and David Chiang, “The Transformer Cookbook”, *Transactions on Machine Learning Research (TMLR)*, 2026.
- 3 Helen Jin*, **Anton Xue***, Weiqiu You, Surbhi Goel, and Eric Wong, “Probabilistic Stability Guarantees for Feature Attributions”, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- 4 Weiqiu You, **Anton Xue**, Shreya Havaladar, Delip Rao, Helen Jin, Chris Callison-Burch, and Eric Wong, “Probabilistic Soundness Guarantees in LLM Reasoning Chains”, in *Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

- 5 Helen Jin*, Shreya Havaldar*, Chaehyeon Kim*, **Anton Xue***, Weiqiu You*, Helen Qu, Marco Gatti, Daniel A Hashimoto, Bhuvnesh Jain, Amin Madani, *et al.*, “The FIX Benchmark: Extracting Features Interpretable to eXperts”, *Journal of Data-centric Machine Learning Research (DMLR)*, 2025.
- 6 Thomas Zhang*, Behrad Moniri*, Ansh Nagwekar, Faraz Rahman, **Anton Xue**, Hamed Hassani, and Nikolai Matni, “On the Concurrence of Layer-wise Preconditioning Methods and Provable Feature Learning”, in *International Conference on Machine Learning (ICML)*, 2025.
- 7 **Anton Xue***, Avishree Khare*, Rajeev Alur, Surbhi Goel, and Eric Wong, “Logicbreaks: A Framework for Understanding Subversion of Rule-based Inference”, in *International Conference on Learning Representations (ICLR)*, 2025.
- 8 **Anton Xue**, Rajeev Alur, and Eric Wong, “Stability Guarantees for Feature Attributions with Multiplicative Smoothing”, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- 9 Xiayan Ji*, **Anton Xue***, Eric Wong, Oleg Sokolsky, and Insup Lee, “AR-Pro: Anomaly Explanation and Repair with Formal Properties”, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- 10 Chang Zhu, Ziyang Li, **Anton Xue**, Ati Priya Bajaj, Wil Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, Adam Doupé, *et al.*, “TYGR: Type Inference on Stripped Binaries using Graph Neural Networks”, in *USENIX Security*, 2024.
- 11 **Anton Xue**, Lars Lindemann, and Rajeev Alur, “Chordal Sparsity for SDP-based Neural Network Verification”, *Automatica*, 2024.
- 12 **Anton Xue**, Lars Lindemann, Alexander Robey, Hamed Hassani, George J Pappas, and Rajeev Alur, “Chordal Sparsity for Lipschitz Constant Estimation of Deep Neural Networks”, in *Conference on Decision and Control (CDC)*, 2022.
- 13 Osbert Bastani, Shuo Li, and **Anton Xue**, “Safe Reinforcement Learning via Statistical Model Predictive Shielding”, in *Robotics: Science and Systems (RSS)*, 2021.
- 14 Rajeev Alur, Phillip Hilliard, Zachary G Ives, Konstantinos Kallas, Konstantinos Mamouras, Filip Niksic, Caleb Stanford, Val Tannen, and **Anton Xue**, “Synchronization Schemas”, in *Symposium on Principles of Database Systems (PODS)*, 2021.
- 15 **Anton Xue** and Nikolai Matni, “Data-driven System Level Synthesis”, in *Learning for Dynamics and Control (L4DC)*, 2021.
- 16 Kedar S Namjoshi and **Anton Xue**, “A Self-certifying Compilation Framework for WebAssembly”, in *Verification, Model Checking, and Abstract Interpretation (VMCAI)*, 2021.
- 17 William T Hallahan, **Anton Xue**, and Ruzica Piskac, “G2Q: Haskell Constraint Solving”, in *International Symposium on Haskell*, 2019.
- 18 William T Hallahan, **Anton Xue**, Maxwell Troy Bland, Ranjit Jhala, and Ruzica Piskac, “Lazy Counterfactual Symbolic Execution”, in *Programming Language Design and Implementation (PLDI)*, 2019.

Awards

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| 2025 | ◇ Postdoctoral fellowship with the Institute for Foundations of Machine Learning at UT Austin |
| 2023 | ◇ Amazon Web Services grant to the University of Pennsylvania ASSET fund |
| 2019 | ◇ University of Pennsylvania ENIAC Fellowship |
| 2019 | ◇ Yale Computer Science Award |
| 2019 | ◇ National Science Foundation Graduate Research Fellowship |
| 2016 | ◇ Yale College Freshman Summer Research Fellowship |

Teaching Experience

- Spring 2023 ♦ **Convex Optimization**
UPenn ESE 605, TA
- 2020 – 2021 ♦ **Fundamentals of Linear Algebra and Optimization**
UPenn CIS 515, TA, Fall 2020, Spring 2021
- Summer 2020 ♦ **Mathematical Foundations of Computer Science**
UPenn CIS 160, TA
- Spring 2019 ♦ **Real Analysis**
Yale MATH 305, Course Grader
- Spring 2018 ♦ **Intensive Algorithms**
Yale CPSC 366, TA
- Spring 2017 ♦ **Design and Analysis of Algorithms**
Yale CPSC 365, TA
- 2016 – 2018 ♦ **Mathematical Tools for Computer Science**
Yale CPSC 202, TA, Fall 2016, Fall 2017, Fall 2018

Projects Supervised

- 2023 – Present ♦ **David Zhang** (UPenn Undergraduate)
Topic: Data Selection for Instruction Tuning
- 2023 – 2024 ♦ **Faraz Rahman** (UPenn Undergraduate)
Topic: Explainability and Interpretability of Diffusion Models
- 2023 ♦ **Jawad Ahmad** (Friends Select School, Philadelphia)
Topic: Programmer-Friendly Code Synthesis with ChatGPT

Research Community Service

- 2024 – Present ♦ **NeurIPS, ICLR, ICML, AAAI**
Reviewer
- 2023 – Present ♦ **SATA, SciForDL, AdvML**
Reviewer
- 2021 – Present ♦ **IEEE L-CSS, CDC, ACC, Automatica, L4DC**
Reviewer
- 2023 ♦ **Tools and Algorithms for the Construction and Analysis of Systems (TACAS)**
Artifact Evaluation Committee
- 2022 ♦ **Principles of Programming Languages (POPL)**
Student Volunteer
- 2021 ♦ **Static Analysis Symposium (SAS)**
Artifact Evaluation Committee
- 2021 ♦ **Conference on Verification, Model Checking, and Abstract Interpretation (VM-CAI)**
Artifact Evaluation Committee
- 2019 – 2021 ♦ **Programming Language Design and Implementation (PLDI)**
Student Volunteer (2019), Artifact Evaluation Committee (2020, 2021)
- 2017, 2019 ♦ **Computer Aided Verification (CAV)**
Student Volunteer

Research Community Service (continued)

- 2017 – 2018 ♦ **Yale University Computer Science Department**
Department Student Advisory Committee

Internship Experience

- 2022 ♦ **SRI International**
Summer Research Intern. Manager: Susmit Jha
Designed and implemented sum-of-squares-based convex optimization methods for verifying neural networks.
- 2019 ♦ **Nokia Bell Labs**
Summer Research Intern. Manager: Kedar Namjoshi
Developed a self-certifying compiler framework for WebAssembly. Paper accepted at the Conference on Verification, Model Checking, and Abstract Interpretation, 2021.
- 2018 ♦ **Harvard University School of Engineering and Applied Sciences**
Summer Research Intern. Advisor: Stephen Chong
Formalized execution semantics of the R programming language. Workshop paper accepted at Formal Methods in Computer-Aided Design, 2018.
- 2017 ♦ **Max Planck Institute for Software Systems**
Summer Research Intern. Advisors: Rupak Majumdar and Damien Zufferey
Applied model checking to 3D printer firmware and contributed to the dReal SMT solver.
- 2016 – 2019 ♦ **Yale University Computer Science Department**
Undergraduate Research Assistant. Advisor: Ruzica Piskac
Developed a symbolic execution engine for Haskell. Paper accepted at Programming Language Design and Implementation, 2019.
- 2014 – 2015 ♦ **Harvard Medical School**
Software Engineering Intern. Advisor: Nils Gehlenborg
Developed visualization software with D3 and contributed to the Refinery Platform for web-based data visualization.
- 2013 – 2014 ♦ **Vertex Pharmaceuticals**
Summer Software Engineering Intern. Manager: Jason Yuen
Developed an inventory scanner and manager for iOS.
Developed a bug-tracking database using Google Web Toolkit.